

Logistic Regression

Question 1: What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:

Simple Linear Regression (SLR) is a statistical method used to model the relationship between a single independent variable (X) and a single dependent variable (Y).

The primary **purpose** of SLR is twofold:

1. **Prediction:** To predict the value of the dependent variable based on a given value of the independent variable.¹
2. **Relationship Analysis:** To determine the strength and direction of the association between two variables (e.g., does increasing X lead to an increase or decrease in Y ?).

Question 2: What are the key assumptions of Simple Linear Regression?

Answer:

For the results of a linear regression model to be reliable, several key assumptions must hold:

- **Linearity:** The relationship between the independent and dependent variables must be linear.²
- **Independence:** The observations in the dataset must be independent of each other.³
- **Homoscedasticity:** The variance of residual errors should be constant across all levels of the independent variable.⁴
- **Normality:** The residuals (errors) of the model should follow a normal distribution.⁵
- **No Outliers:** Significant outliers can disproportionately influence the slope of the regression line.⁶

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer:

The standard mathematical equation for a simple linear regression model is:

Logistic Regression

$$y = \beta_0 + \beta_1 x + \epsilon$$

Explanation of terms:

- y : The **Dependent Variable** (Target) we are trying to predict.⁷
- x : The **Independent Variable** (Predictor).⁸
- β_0 : The **Y-intercept**, representing the value of y when $x = 0$.¹¹
- β_1 : The **Slope** (Coefficient), representing the change in y for every one-unit change in x .
- ϵ : The **Random Error term**, which accounts for the variation in y that cannot be explained by the linear relationship with x .¹²

Question 4: Provide a real-world example where simple linear regression can be applied.

Answer:

A classic example is predicting House Prices based on Square Footage.¹⁵

In this scenario:

- **Independent Variable (X):** The size of the house in square feet.
 - **Dependent Variable (Y):** The selling price of the house.
- By applying SLR, a real estate agent can estimate how much the price of a home will increase for every additional square foot of space added.¹⁶

Question 5: What is the method of least squares in linear regression?

Answer:

The Method of Least Squares is a mathematical optimization technique used to find the "best-fit" line for a set of data points.¹⁷ It works by minimizing the sum of the squares of the vertical deviations (residuals) between each observed data point and the fitted line. By squaring the errors, the method ensures that positive and negative deviations don't cancel each other out and gives more weight to larger errors.¹⁸⁺¹

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:

Logistic Regression

Logistic Regression is a supervised learning algorithm used for classification tasks, despite its name.¹⁹ It predicts the probability of a categorical dependent variable (usually binary, like Yes/No).

Key Differences:

| Feature | Linear Regression | Logistic Regression |

| :--- | :--- | :--- |

| Output | Continuous value (e.g., price, height) | Probability/Category (e.g., 0 or 1) |

| Equation | Straight line ($y = mx + c$) | Sigmoid/Logistic function (S-curve) |

| Goal | Minimize Mean Squared Error | Maximize Likelihood of observed classes |

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Answer:

1. **Mean Absolute Error (MAE):** The average of the absolute differences between predicted and actual values.²⁰ It is easy to interpret as it is in the same units as the target variable.
2. **Mean Squared Error (MSE):** The average of the squared differences between predicted and actual values.²¹ It penalizes larger errors more heavily than smaller ones.²²
3. **Root Mean Squared Error (RMSE):** The square root of MSE.²³ It provides an error metric in the same units as the dependent variable while maintaining the penalty for large errors.

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:

R-squared (R^2), also known as the Coefficient of Determination, measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s).²⁵

- An R^2 of 0 means the model explains none of the variability.²⁷

Logistic Regression

- An R^2 of 1 means the model explains all the variability.²⁹ It essentially tells you how well your data "fits" the regression model.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept.

Answer:

```
Python
import numpy as np
from sklearn.linear_model import LinearRegression

# Sample data: X (Independent), y (Dependent)
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 5, 4, 5])

# Initialize and fit the model
model = LinearRegression()
model.fit(X, y)

# Retrieve coefficients
slope = model.coef_[0]
intercept = model.intercept_

print(f"Slope (beta_1): {slope}")
print(f"Intercept (beta_0): {intercept}")

# Output:
# Slope (beta_1): 0.6
# Intercept (beta_0): 2.2
```

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

The coefficients provide the specific "rules" for the relationship:

1. **The Intercept (β_0):** This is the expected mean value of Y when $X = 0$.³⁰ In some contexts (like house prices), this may just be a baseline value; in others, it may not have a practical physical meaning if $X=0$ is impossible.³¹³²³³

Logistic Regression

2. **The Slope** (³⁴ β_1): This indicates the magnitude and direction of change.³⁵ For every **one-unit increase** in the independent variable (³⁶ X), the dependent variable (³⁷ Y) is expected to change by the value of ³⁸ β_1 (increasing if positive, decreasing if negative).³⁹

+1